

The Banach-lattice-algebra representation problem

Literature-status packet for arXiv:2407.09149

literature_already_answered—negative

Source question

The source is David Muñoz-Lahoz, *Band projections and order idempotents in Banach lattice algebras*, arXiv:2407.09149v2 (2024). Question 1.1 on page 2 (quoting Wickstead’s Question 5.3) asks:

Is every Banach lattice algebra isometrically (isomorphic) to a closed subalgebra and sublattice of some space of regular operators?

Here the intended concrete targets are $\mathcal{L}^r(X)$ for Dedekind complete Banach lattices (or vector lattices) X , and a representation preserves both lattice and algebra operations.

Separate later answer

The same author subsequently answered this question explicitly:

David Muñoz-Lahoz, *Wickstead’s conjecture on positive projections and non-representable Banach lattice algebras*, arXiv:2604.14697v3, 10 August 2026.

Its abstract states that there are two-dimensional Banach lattice algebras with no faithful representation as regular operators on any Dedekind complete Banach lattice. The introduction, page 2, says verbatim that this “answers [9, Question 5.3] and [10, Problem 2] in the negative.” Thus the supporting authors—indeed, the source and supporting author are the same—explicitly identify the resolved problem.

The decisive result is Proposition 4.3 on pages 14–15. If a lattice-ordered algebra contains nonzero positive idempotents e, p with $ep = pe = p$ and

$$p = \alpha e + x, \quad x \wedge e = 0, \quad \alpha \notin \{0\} \cup \{1/n : n \in \mathbb{N}\},$$

then it has no faithful representation on $\mathcal{L}^r(X)$ for any Dedekind complete vector lattice X .

Example 4.4, pages 15–16, applies this to $\mathbb{R}^2$ with pointwise multiplication and cone

$$P_\beta^1 = \{(s, t) \in \mathbb{R}^2 : 0 \leq s, \beta s \leq t \leq s\}, \quad -1 \leq \beta \leq 0.$$

This is normable as a unital Banach lattice algebra. Whenever

$$\beta \notin \{0\} \cup \{-1/(n-1) : n \in \mathbb{N}, n \geq 2\},$$

it admits no faithful representation, whether or not the representation is required to be unital. For example, $\beta = -2/3$ gives a concrete two-dimensional counterexample.

Status and scope

The answer to Question 1.1 of arXiv:2407.09149 is therefore **no**. This is an exact later-paper resolution, not a new result of the present run. The later theorem is stronger than merely disproving isometric representability: it rules out every faithful lattice-and-algebra representation of the displayed examples on any Dedekind complete vector lattice.

Search evidence: exact-title and exact-question searches on 11 August 2026 found arXiv:2604.14697v3; its abstract, introduction, Proposition 4.3, and Example 4.4 were checked against the local supporting PDF. No ambiguity remains about the identification.

References

- [1] D. Muñoz-Lahoz, *Band projections and order idempotents in Banach lattice algebras*, J. Math. Anal. Appl. 543 (2025), 129045; arXiv:2407.09149.
- [2] D. Muñoz-Lahoz, *Wickstead’s conjecture on positive projections and non-representable Banach lattice algebras*, arXiv:2604.14697v3 (2026).